

SQL Homework

The DataCamp course Introduction to SQL provides a baseline knowledge of the structured query language. Complete the DataCamp course. Follow the directions posted to Week 6 to install MySQL Workbench and to import the stda database.

Complete the following prompts and submit a PDF or word document with your responses. Some of the prompts require commands not covered in the DataCamp course, but are readily found online. I will be testing your ability to self teach several commands. I recommend using www.w3schools.com as your primary resource, although you may prefer others.

Part 1: Comprehension

1. (10 pts) Based on the DataCamp activity “Introduction to SQL,” what are the advantages of SQL over spreadsheets?

SQL databases are more powerful, and they are safer to use because they are encrypted.

2. (10 pts) Based on the DataCamp activity “Introduction to SQL,” what are keywords in SQL? Are they lowercase or capital, and why?

Keywords are reserved words used to indicate what operation we would like our code to perform. They are all capital to distinguish from fields.

3. (10 pts) Based on the DataCamp activity “Introduction to SQL,” what is a field in SQL? Are they lowercase or capital, and why?

A field is a column name. They are written in lowercase to distinguish from capitalized SQL words to make queries easier to understand and easier to read.

Part 2: Implementation

1. (10 pts) The stda database has a table named orders, which can be referenced as `stda.orders`.

Create a new view called `my_view`. `my_view` should select all fields from `stda.orders` where the segments is equal to “Home Office”. Provide the script that generates the specified view.

https://www.w3schools.com/sql/sql_view.asp

<https://www.youtube.com/watch?v=vLLkNI-vkV8>

```
CREATE VIEW my_view AS
```

```
SELECT *
```

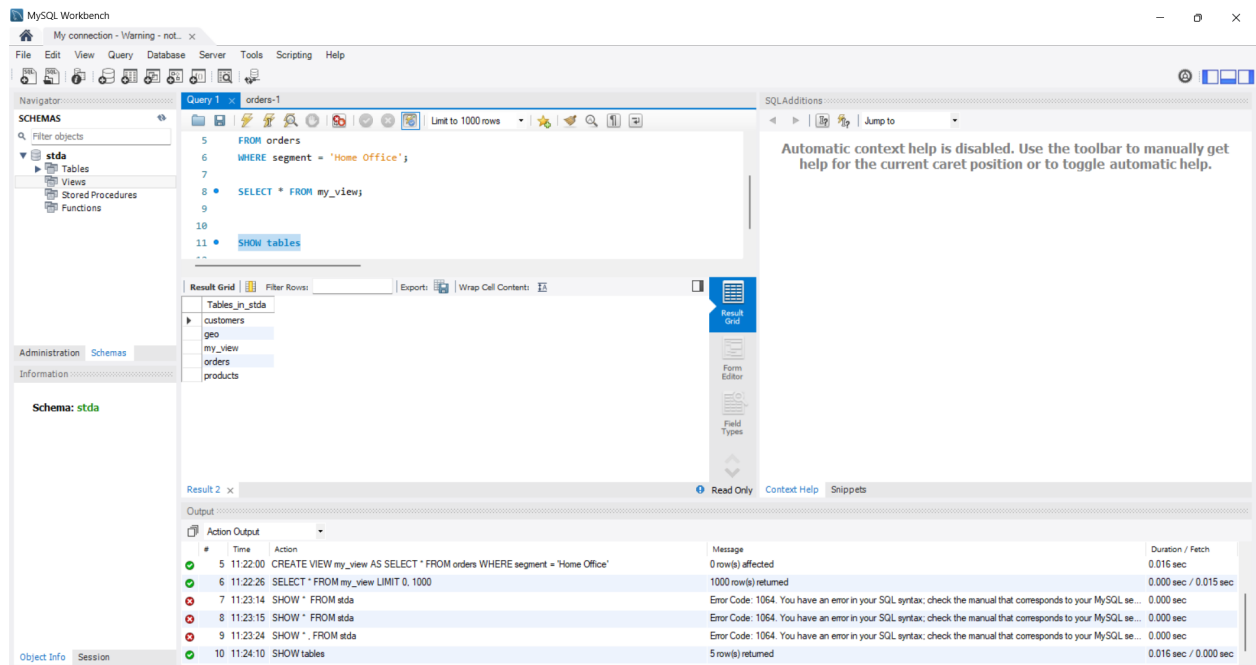
```
FROM orders
```

```
WHERE segment = 'Home Office';
```

2. (10 pts) Provide a line of code that will **SHOW** all the **tables** in the current schema.

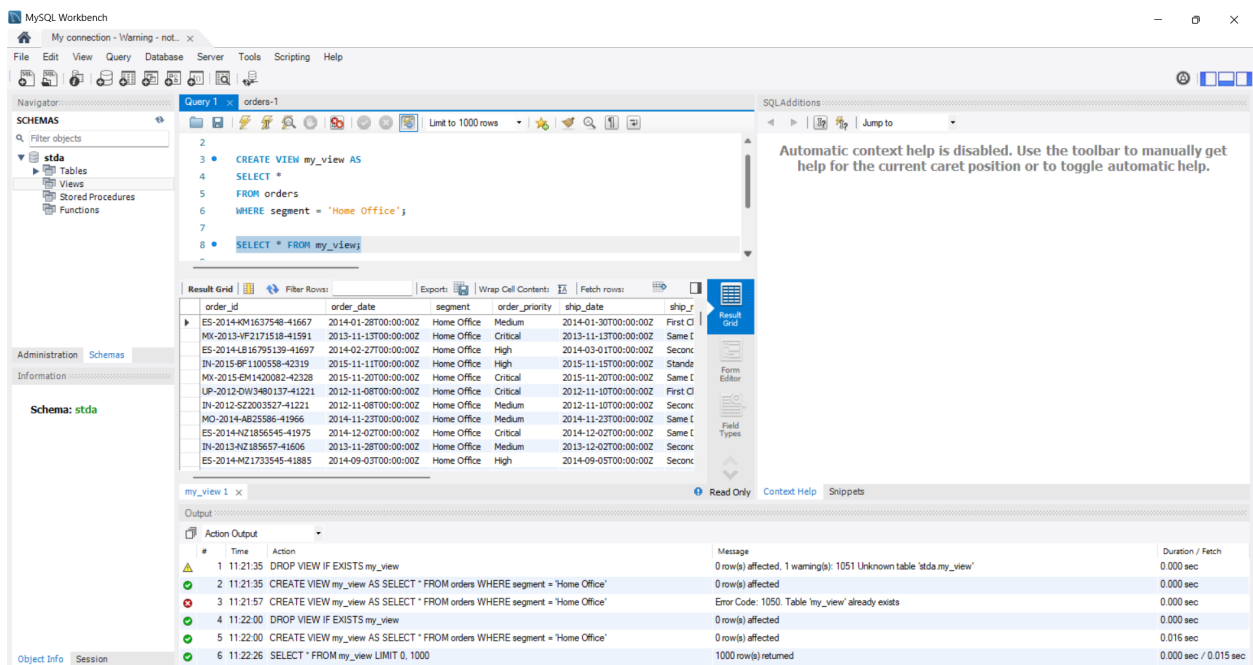
Screenshot the results.

SHOW Tables

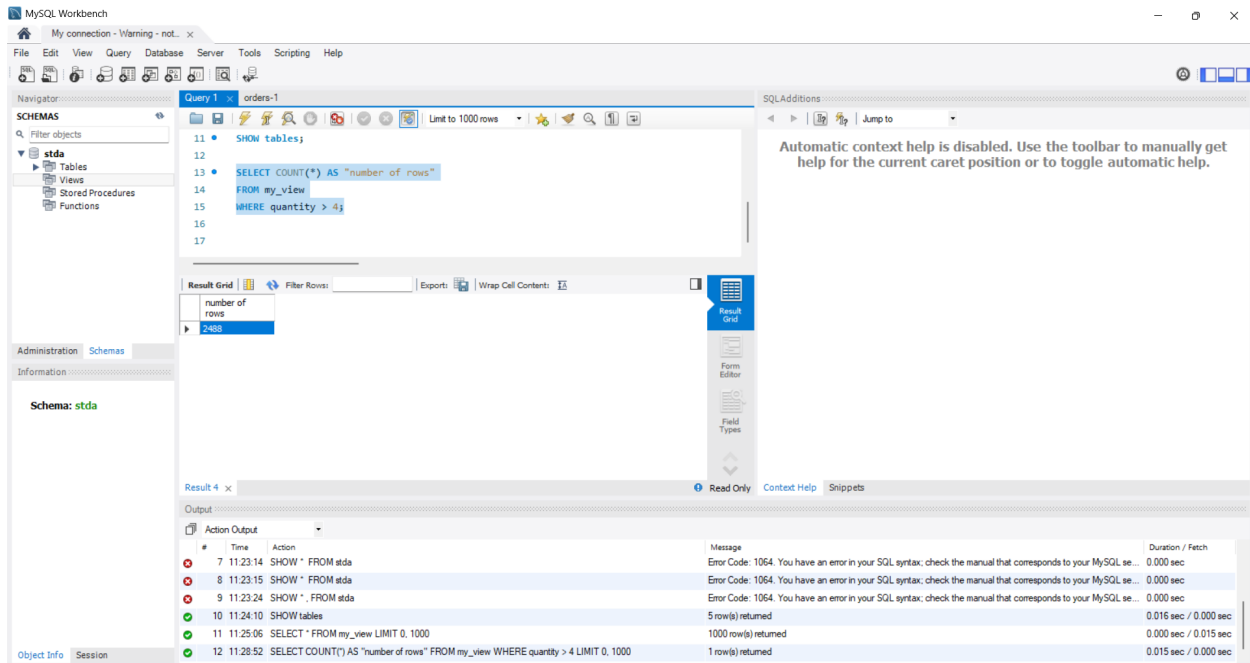


3. (10 pts) Provide a line of code that will **SELECT** all the fields **FROM** **my_view**

Screenshot the results.



4. (10 pts) **SELECT** the **COUNT** of records in all the fields **as** a result called "number of rows" **WHERE** the field "quantity" is greater than 4.



Screenshot the results.

<https://www.w3resource.com/sql/aggregate-functions/count-function.php>

Part 3: Self Teaching

- (10 pts) Discuss a specific business use case of joining tables in this database.

An specific business scenario you may want to join tables in this database is if you need to see what customers have ordered. Then from this you are able to pull up what customers have ordered previously. We can do this by combining the two tables based on the customer_id columns

- (10 pts) Provide the code to join the suggested tables.

SELECT *

FROM customers

INNER JOIN orders

ON customers.customer_id = orders.customer_id;

- (10 pts) Screenshot the resulting table of the join statement.

